E.G.S.PILLAYARTS&SCIENCECOLLEGE(AUTONOMOUS)
NAGAPATTINAM–611002.TAMILNADU
(Accredited by NAAC With "A" Grade)
(MBA–ApprovedbyAICTE,NewDelhi)
(AffiliatedtoBharathidasanUniversity,Tiruchirappalli–24)

DEPARTMENT OF BIOTECHNOLOGY

Choice Based Credit System – Learning Outcomes Based Curriculum Framework (CBCS – LOCF)

(Applicable to the candidate admitted from the academic year 2026 – 2027 onwards)

EDAYATHANGUDY G.S. PILLAY ARTS & SCIENCE COLLEGE
An Autonomous Institution, Affiliated to Bharathidasan University,
Tiruchirappalli Nagore Post,
Nagapattinam – 611 002, Tamil Nadu
B.SC. BIOTECHNOLOGY CURRICULUM STRUCTURE
(Applicable to the candidate admitted from the Academic year 2025-2026 onwards)

Sem	Course	Title	Ins. Hrs/ Week	Credits	Exam. Hrs.	Marks		Total
						Int.	Ext	
I	Language Course – I (LC) - Tamil/other languages	Tamil	6	3	3	25	75	100
	English Course- I (ELC)	English	6	3	3	25	75	100
	Core Course – I (CC)	Cell Biology	5	5	3	25	75	100
	Core Practical – I (CP)	Cell Biology (P)	4	4	3	50	50	100
	Allied Course – I (AC)	Biochemistry	4	4	3	25	75	100
	Allied Practical – I(AP)	Biochemistry (P)	3	3	3	50	50	100
	Value Education	Value Education	2	2	3	25	75	100
	Total		**30**	**24**	-	-	-	**700**
II	Language course – II (LC) - Tamil/other languages	Tamil	6	3	3	25	75	100
	English Course- II (ELC)	English	6	3	3	25	75	100
	Core Course – II (CC)	Microbiology	5	5	3	25	75	100
	Core Practical – II (CP)	Microbiology (P)	4	4	3	50	50	100
	Allied Course – II (AC)	Plant and Animal Physiology	4	4	3	25	75	100
	Allied Practical-II (AP)	Plant and Animal Physiology (P)	3	2	3	50	50	100
	Environmental Studies	Environmental Studies	2	2	3	25	75	100
	Total		**30**	**23**	-	-	-	**700**
III	Language course – III (LC) - Tamil/other languages	Tamil	6	3	3	25	75	100
	English Course- III(ELC)	English	6	3	3	25	75	100
	Core Course – III (CC)	Molecular Biology and Genetics	5	5	3	25	75	100
	Core Practical – III (CP)	Molecular Biology and Genetics (P)	4	4	3	50	50	100
	Allied Course – III (AC)	Bioinstrumentation	4	4	3	25	75	100
	Allied Practical – III(AP)	Bioinstrumentation (P)	3	2	3	50	50	100
	Non Major Elective - I	A. Food and Nutritional Biotechnology B. Herbal Biotechnology	2	2	3	25	75	100
	Part IV	Health and Wellness	-	1	-	-	-	100
	Total		**30**	**24**	-	-	-	**800**

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IV	Language course – IV(LC) - Tamil/other languages	Tamil	6	3	3	25	75	100
	English Course- IV(ELC)	English	6	3	3	25	75	100
	Core Course – IV (CC)	Immunology	5	5	3	25	75	100
	Core Practical – IV(CP)	Immunology (P)	4	4	3	50	50	100
	Allied Course – IV (AC)	Bioinformatics	4	3	3	25	75	100
	Allied Practical – IV(AP)	Bio informatics (P)	3	2	3	50	50	100
	Non Major Elective - II	A.Biotechnology for Human Health B.Dairy Biotechnology	2	2	3	25	75	100
	Total		**30**	**22**	-	-	-	**700**
V	Core Course – V (CC)	Gene Manipulation Technology	6	5	3	25	75	100
	Core Course – VI (CC)	Plant and Animal Biotechnology	6	5	3	25	75	100
	Core Course – VII (CC)	Microbial and Dairy Biotechnology	5	5	3	25	75	100
	Core Practical – V(CP)	Gene Manipulation Technology, Plant and Animal Biotechnology, Microbial and Dairy Biotechnology	5	4	3	50	50	100
	Major based Elective –I	A. Pharmaceutical Biotechnology B. Medical Biotechnology	4	4	3	25	75	100
	Skill Based Elective - I	A. Medical Lab Technology B. Medical Coding	2	2	3	25	75	100
	Soft Skill Development	Soft skill development	2	2	3	25	75	100
	Internship	Internship	*	2	-	20	80	100
	Total		**30**	**29**	-	-	-	**800**
VI	Core Course – VIII (CC)	Environmental Biotechnology	6	5	3	25	75	100
	Core Course – X (CC)	Biosafety, Bio ethics and IPR	6	5	3	25	75	100
	Core Practical – VI (CP)	Environmental Biotechnology	4	4	4	50	50	100
	Major Based Elective –II	A. Nano Biotechnology B. Food Biotechnology	5	4	3	25	75	100
	Skill Based Elective II	A. Molecular Diagnostics B. Stem Cell technology	4	2	3	25	75	100
	Project	Dissertation and Viva-voce	4	3	-	20	80	100
	Extension Activities**	Extension Activities	-	1	-	-	-	-
	Gender Studies	Gender studies	1	1	3	25	75	100
	Total		**30**	**25**	-	-	-	**700**
	Grand Total		**180**	**147**				**4400**

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SUMMARY OF CURRICULUM STRUCTURE OF UG PROGRAMMES

SL. NO	Part	Type of Courses	No. Of Course	No. of. Credits	Marks
1.	I	Language Courses	4	12	400
2.	II	English Courses	4	12	400
3.	III	Core Courses	9	45	900
4.		Core Practical	6	24	600
5.		Allied Course 1 & 11	4	16	400
6.		Allied Practical	4	8	400
7.		Major Based Elective Courses	2	8	200
8.		Project	1	3	100
9.		Internship	1	2	100
10.	IV	Health Wellness	1	1	100
11.		Non-Major Elective Courses	2	4	200
12.		Skill Based Elective Courses	2	4	200
13.		Soft Skills Development	1	2	100
14.		Value Education	1	2	100
15.		Environmental Studies	1	2	100
16.	V	Gender Studies	1	1	100
17.		Extension Activities	1	1	0
Total			**45**	**147**	**4400**

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SEMESTER I

First Year	CORE COURSE – I (CC)	Credit : 5
Semester - I	CELL BIOLOGY	Sub Code:25BTCC1

(Theory)

LEARNING OBJECTIVES

- Familiarize the student in different areas of cell biology, including cell structure and functions and its organelles.
- Develop comprehensive understanding of the cellular membranes and matrices.
- Impact the knowledge of microscopic techniques to understand the cell structure

UNIT	CONTENTS	HOURS
I	Basic & History of cell Biology, cell as basic unit of life, Discovery and diversity of cells - Cell theory - Structure of prokaryotic (bacteria) and eukaryotic cells (plant and animal cells).	10
II	Structure and function of cell wall – Bacteria, Animal and Plant. Ultra structure of plasma membrane – fluid mosaic model, membrane fluidity, Transport across membranes - Symport, antiport, uniport, active and passive transport.	15
III	Structure & Functions of cell organelles: Endoplasmic Reticulum (SER & RER), Golgi apparatus, lysosomes, microbodies (peroxisomes and glyoxysomes), ribosomes and its types, centrioles, basal bodies. Structure and functions of mitochondria and chloroplast.	15
IV	Structure and organization of nucleus, nuclear membrane, organization of chromosomes-structural organization of chromatids, centromere, chromatin, telomere, nucleosomes, euchromatin and heterochromatin, specialized structures-polytene and lampbrush chromosome	10
V	Cell cycle - Cell cycle checkpoints - Cell division - Mitosis and Meiosis - Cellular differentiation - Cell junctions - Cell Adhesion – Extracellular Matrix - Cell to cell communications - Signal transduction - G - Protein Coupled Receptors - Signal transduction pathways	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic concepts of cell biology, cell structure, and its importance in living systems	K1
CO2	Describe the structure and functions of cellular organelles and biomolecules involved in cellular activities	K2
CO3	Explain various cellular processes such as membrane transport, cell signalling, and cell division	K2
CO4	Analyze the structure – function relationships of cells and compare different processes involved in growth and metabolism	K4
CO5	Demonstrate laboratory techniques used in cell biology and interpret observations related to cell structure and function.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Text Books

- Alberts, B., Johnson, A., Lewis, J., Raff, M., Roberts, K. & Walter, P. (2014). Molecular Biology of the Cell-6th Edition.Garland Publishing (Taylor & Francis Group)
- Geoffrey Cooper. (2018). The Cell: A Molecular Approach, 8th Edition. Oxford University press.
- Gerald Karp, Janet Iwasa & Wallace Marshall. (2019). Karp's Cell and Molecular Biology-9th Edition. John Wiley & Sons. Inc.
- Harvey Lodish. (2014). Molecular Cell Biology, 7th Edition, W.H.Freeman and Company.
- Jeff Hardin, Greg Bertoni & Lewis. J. Kleinsmith. (2016). Becker's World of the Cell, 9th Edition. Pearson Benjamin Cummings Publication.
- E.D. P. De Robertis and E.M.F. De Robertis, Jr. 2012 Cell and Molecular Biology (Eighth edition). B.I. Waverly Pvt.Ltd. New Delhi.
- Harvey Lodish, Arnold Berk, S. Lawrence Zipursky, Paul Matsudaira, David Baltimore and James Dernell, 2009. Molecular Cell Biology (Fourth Edition). Media Connected –W.H.Freeman and Company.
- P.S. Verma and V.K. Agarwal, 2012, Concepts of Cell Biology. S.Chand & Company Ltd., New Delhi

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First Year	CORE COURSE – I (CP)	Credit : 4
Semester - I	CELL BIOLOGY	Sub Code: 25BTCP1

(Practical)

LEARNING OBJECTIVES

- Demonstrate the operation of Light Microscope
- Identify of macromolecules
- Isolate and identify plant, and animal cells.
- Develop skill to perform cell fractionations.

PRACTICALS

1. Components of a Compound / Light Microscope.
2. Structure observation of Prokaryotic cells
3. Structure observation of Eukaryotic cell
4. Plasmolysis in onion
5. Mitosis in onion root tip.
6. Meiosis in flower buds of onion
7. Cell counting method- plant cells-stomatal density
8. Osmosis experiment using potato
9. Cell fractionation and Identification of cell organelles (Demo)

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Identify and understand the basic laboratory instruments, safety measures, and techniques used in cell biology practicals	K1
CO2	Describe the structure and organization of cells using microscopeic observation and staining techniques	K2
CO3	Perform basic cell biology experiments such as preparation of slides, staining methods, and identification of cellular components	K2

CO4	Analyze experimental observations and interpret the structure and functions of cellular components	**K4**
CO5	Demonstrate practical skills in handling laboratory equipments, maintaining records and applying cell biology techniques for experimental studies	**K3**
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

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First Year	ALLIED COURSE – I (AC)	Credit : 5
Semester - I	BIOCHEMISTRY	Sub Code:25BCAC1

(Theory)

Objectives

- Demonstrates the formation of different types of solutions, concentrations of solution sand preparation of buffer solutions
- Recall the Structure, Classification, Chemistry and Properties of Carbohydrates and Explain Various Biochemical Cycles involved in Carbohydrate Metabolism.
- Recall the Structure, Classification, Chemistry and Properties of Lipids, Nucleic acid and Explain Various Biochemical Cycles involved in Fatty acid and Nucleic acid Metabolism.

UNIT	CONTENTS	HOURS
I	Acids & Bases properties and differences, concepts of acids and bases- Arrhenius, Lowry-Bronsted and Lewis. Concentration of solution, ways of expressing concentrations of solutions-percent by weight, normality, molarity, molality, mole fractions, pH of solution, pH scale, measurement of pH, Buffer solutions, properties of buffers, Henderson-Hasselbalch equation, mechanism of buffer action of acidic buffer and basic buffer	10
II	Carbohydrates: Classification. Monosaccharides – D and L designation, open chain and cyclic structures, epimers, anomers and mutarotation. Occurrence, structure and biological importance of disaccharides (sucrose, lactose, maltose) and polysaccharides (storage-starch, glycogen; structural – cellulose). Carbohydrate Metabolism: Glycolysis, TCA cycle	15
III	Proteins: Structure, Classification, Physical and Chemical properties of amino acids. Essential and non-essential aminoacids. Biological importance of proteins; Classification based on structure and functions, structural organization of proteins (primary, secondary, tertiary and quaternary structures). Shikimate pathway of amino acids biosynthesis	15
IV	Classification of Lipids. Characteristics, Properties and Biological importance of lipids. Metabolism of Fatty acids, triglycerides, phospholipids, cholesterol. B-oxidation of fatty acids. Classification of nucleic acids. Purine and Pyrimidine	10

	bases. Classification of DNA & RNA. Metabolism of Nucleic acids, Salvage pathway	
V	Metabolism: Catabolism & Anabolism. Energy metabolic pathways – Glycolysis, Krebs cycle, Entner-Doudoroff -pathway, Gluconeogenesis, pentose phosphate pathway. C3, C4 cycle of CO2 reduction, Photo phosphorylation & Oxidative phosphorylation. Urea cycle. Denovo synthesis of nucleotide biosynthesis.	**10**

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts of biochemistry and the structure and functions of biomolecules	**K1**
CO2	Describe the metabolism and biological significance of carbohydrates, proteins, lipids, and nucleic acids.	**K2**
CO3	Explain the role of enzymes, vitamins, and hormones in various biochemical processes	**K2**
CO4	Analyze metabolic pathways and biochemical reactions involved in cellular functions and energy production	**K4**
CO5	Demonstrate the application of biochemical principles and laboratory techniques in biological and clinical studies	**K3**
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

References and Textbooks:

- Lehninger (2013) Principles of Biochemistrty 4 th edition WH Freeman and Company NY
- Murray et al., (2003) Harper's biochemistry 26 th edition Appleton and Lange Publishers Florida USA
- Geoffrey L. Zubay, William W. Parson, Dennis E. Vance, 1995, Principles of Biochemistry, W.C. Brown Publishers, 1995, 3rd edition.
- LubertStryer (2007) Biochemistry –Stanford University 5 th Edition-W H Freemann and company San Francisco
- BahlArun, Bahl B. S. (2016), A Textbook of Organic Chemistry, 22nd Edition, S. Chand & Sons publications

- P.L. Soni , A Text-book of Inorganic Chemistry, 11th Edition, S. Chand & Sons publications
- AbhilashaShourie, Shilpa S, Chapadgoankar&Anamika Singh (2020) Textbook of Biochemistry 1st Edition
- J.L. Jain, 2016, Fundamentals of Biochemistry, S. Chand publication, 7th edition.
- A.C. Deb, 2016, Fundamentals of Biochemistry, New central book agencies, 7th edition.
- Satyanarayana .U, 2016, Biochemistry, MJ publishers 3rd edition (2006).

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First Year	ALLIED COURSE – I (AP)	Credit : 4
Semester - I	BIOCHEMISTRY	Sub Code:25BCAP1

(Practical)

Objectives:

- Qualitatively analyze the carbohydrates and amino acids and report the type of carbohydrate based on specific tests. Differentiate the carbohydrates based microscopic examination of the crystal.
- Understand the methods of acidimetry, alkalimetry and permanganometry.
- Quantify Ascorbic acid in lemon by Dichlorophenol indo phenol dye method, Glycine by sorensons formal titration method.
- Estimate Glucose,Cholesterol and Proteins.

PRACTICALS

Systematic analysis of Organic compounds

1. Functional group tests (Carboxylic acid (Benzoic acid, phthalic acid), Phenol, Urea, Benzaldehyde, Aniline (Aniline not to be given for exam)
2. Detection of elements (N, Halogens)
3. Distinguish between aliphatic and aromatic compounds.
4. Distinguish between Saturated and unsaturated compounds

Qualitative Analysis

5. Qualitative analysis of carbohydrates - Glucose, Fructose, Lactose, maltose, sucrose, starch & glycogen.
6. Qualitative analysis of amino acids - Tyrosine, Tryptophan, Arginine, Proline and Cysteine.

Volumetric Analysis

7. Estimation of Glycine- Formal Titration.
8. Determination of Ascorbic acid – DCPIP method.
9. Estimation of Ferrous sulphate using standard Mohr's salt

Colorimetric Analysis

10. Estimation of carbohydrate – Anthrone method

11. Estimation of Cholesterol- Zak's method

12. Estimation of proteins – Bradford's method

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Identify and understand laboratory instruments, safety procedures, and basic techniques used in biochemistry practicals	K1
CO2	Describe the qualitative and quantitative analysis of biomolecules such as carbohydrates, proteins, lipids, and enzymes	K2
CO3	Perform biochemical experiments and apply standard laboratory procedures for the analysis of biological samples	K2
CO4	Analyze experimental data and interpret biochemical reactions and results obtained during laboratory investigations	K4
CO5	Demonstrate practical skills in handling laboratory equipments, maintaining records and applying biochemical techniques in research and diagnostic studies	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

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SEMESTER II

First Year	CORE COURSE – II (CC)	Credit : 5
Semester - II	MICROBIOLOGY	Sub Code:25BTCC2

(Theory)

Objectives

- Understand the classification of Microorganisms and structure of bacteria
- Understand the various microbiological techniques, different types of media, and techniques involved in culturing microorganisms.
- Categorize the methods of sterilization and identify the significance of culture media in the growth of different microbes.

UNIT	CONTENTS	HOURS
I	History and Scope of microbiology: Discovery of Microorganisms. General features and Classification of Archaea, Bacteria, Fungi, Algae, Protozoa, Viruses and Prions.	10
II	Structure of bacteria - Bacterial growth and measurement of growth, Media – types and preparation- Pure culture techniques- streak, pour and spread plate technique - staining methods (Gram's, capsule, spore, LCB mount)- methods of preservation and storage of microbes. Culture of fungi, virus and algae.	15
III	Microbial Control: Sterilization, disinfection, antisepsis, fumigation. Physical control: Temperature (moist heat, dry heat, and incinerators), desiccation, osmotic pressure, radiation, UV-light, ultrasonic sound waves, filtration. Chemical control: Antiseptics and disinfectants (halogens, alcohol, gaseous sterilization).	15
IV	Bioinsecticides - Bacillus thuringiensis, Baculoviruses- Biofertilizers - Azospirillum and blue green algae - single cell protein – prebiotics and probiotics - Dairy products (Cheese and Yoghurt).	10
V	Microbial Disease- host -pathogen interaction, clinical features, lab diagnosis and treatment of Airborne disease (Pneumonia, Chicken pox), food borne	10

	disease (Typhoid, Aspergillosis), Water borne disease (Cholera, Amoebiasis), Sexually transmitted disease (AIDS, Trichomoniasis), Vector borne disease (Dengue, Malaria).	

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts of microbiology and the classification of microorganisms	K1
CO2	Describe the structure, characteristics, and growth patterns of microorganisms	K2
CO3	Explain the role of microorganisms in health, industry, environment and biotechnology application	K2
CO4	Analyze microbial growth, metabolism and factors affecting microbial activities	K4
CO5	Demonstrate knowledge of microbiological techniques, aseptic methods, and microbial control measures	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

References and Textbooks:

- M.J. Pelczar Jr. E.C.S. Chan and N.R. Kreig, Microbiology (5th edition), Tata MaCraw-Hill, New Delhi;
- R. Ananthanarayanan. and C.K.Jayaram Panickar, Text book of Microbiology (9th edition), Orient Longman Publications, New Delhi
- Lansing M. Prescott, John. P. Harley, Donald A. Klein, 1999. Microbiology (9th edition) WCB MaCraw-Hill, New York
- Sundararajan S (2003). College Microbiology, revised edition, Vardhana publications, Banglore.
- R.C. Dubey, D.K.Maheswari, A Text book of Microbiology (2005), S.Chand & C7ompany Ltd. New Delhi

- Gerard J. Tortora, Berdell R. Funke, Christine L. Case, Derek Weber & Warner Bair. (2019). Microbiology: An Introduction. 4th Edition. Pearson Benjamin Cummings Publication.
- Gold man, E & Green, H. (2008). Practical handbook of microbiology. CRC press. Gunasekaran, P. (1995). Laboratory Manual in Microbiology. New Delhi: New Age International (P) Ltd. Publishers.
- Jayaraman, J. (1981). Laboratory Manual in Biochemistry. New Delhi: New Age International (Pvt.)
- Ltd. Publishers.
- Joanne Willey, Linda Sherwood & Chris Woolverton (2013). Prescott's Microbiology. Tata McGraw – Hill Publication.

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First Year	CORE ORACTICAL – I (CP)	Credit : 4
Semester - II	MICROBIOLOGY	Sub Code: 25BTCP2

(Practical)

Objective:

- Describe the general Laboratory safety & Sterilization Techniques.
- Develop Skills in Media Preparation, Isolation & Serial Dilution Techniques and Pure Culture Techniques.
- Microscopically analyze the morphological features of Bacteria and fungi and define various Staining Techniques.

PRACTICALS

1. Microscope and its components
2. Simple Staining of Bacteria
3. Gram Staining of Bacteria
4. Preparation of Solid, Liquid, and Semi-Solid Media
5. Pure culture techniques: Streak, Pour and Spread plate technique.
6. Bacterial Motility Assay
7. Microbial Growth Curve Analysis
8. Isolation and enumeration of bacteria from environmental samples such as air, water and soil.
9. Isolation and enumeration of fungi from environmental samples such as air, water and soil.
10. Antibiotic sensitivity test (Demo)

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define basic microbiological laboratory equipment, media, and safety procedures.	K1
CO2	Describe sterilization, disinfection, and aseptic techniques used in microbiology laboratories.	K2
CO3	Demonstrate preparation of culture media, staining techniques, and microbial isolation methods.	K2
CO4	Analyze microbial growth characteristics and identify microorganisms using microscopic and biochemical methods.	K4

CO5	Evaluate sources of microbial contamination and apply appropriate control measures in laboratory practices	**K3**
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

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First Year	ALLIED COURSE – II (AC)	Credit : 5
Semester - II	PLANT AND ANIMAL PHYSIOLOGY	Sub Code:25BCAC2

(Theory)

Objective:

This course aims to provide a comprehensive understanding of the physiological processes in plants and animals, focusing on key mechanisms that sustain life, adaptation, and homeostasis.

UNIT	CONTENTS	HOURS
I	Photosynthesis- Light harvesting complex; Photoprotective mechanisms; CO 2 fixation-C3, C4 and CAM pathways. Sensory photobiology: Structure, function and mechanisms of action of phytochromes, cryptochromes and phototropins; stomatal movement; photoperiodism and biological clocks.	10
II	Uptake, transport and translocation of water, ions, solutes and macromolecules from soil, through cells, across membranes, through xylem and phloem; transpiration; mechanisms of loading and unloading of photoassimilates	15
III	Responses of plants to biotic (pathogen and insects) and abiotic (water, temperature and salt) stresses; mechanisms of resistance to biotic stress and tolerance to abiotic stress.	15
IV	Respiratory system: anatomy and structure transport of gases, exchange of gases, waste elimination, neural and chemical regulation of respiration. Digestive system: Digestion, absorption, energy balance, BMR.	10
V	Excretory system: Comparative physiology of excretion, kidney, 12 urine formation, urine concentration, waste elimination, micturition, regulation of water balance, blood volume, blood pressure, electrolyte balance, acid-base balance. Endocrinology and Reproduction- Endocrine glands, basic mechanism of hormone action, hormone and diseases, neuroendocrine regulation.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic principles of plant and animal physiology and their importance in biological systems.	K1
CO2	Describe physiological processes in plants such as photosynthesis, respiration, transpiration, mineral nutrition, and plant growth regulation.	K2
CO3	Explain physiological functions of animal organ systems including digestion, respiration, circulation, excretion, and nervous coordination	K2
CO4	Analyze the mechanisms regulating growth, development, reproduction, and homeostasis in plants and animals	K4
CO5	Evaluate physiological responses of plants and animals to environmental factors and their adaptive significance	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Textbooks:

1. Frank B Salisbury and Cleon Wross. Plant Physiology, CBS Publishers and Distributors, New Delhi.
2. Malcolm S Wilklins. Advanced Plant Physiology.
3. Pushit SS. Hormonal regulation of plant growth and development.
4. Ganong H. (1989). Review of Medial Physiology, 14th Edition, Appleton &Lange publisher, New York.
5. EcKert R. (2002). Animal Physiology, 5th Edition, W.H.Freeman.

References

1. Sltyar RG. Plant water relationships.
2. Roy G Noggle and George J Friltz. Introductory Plant physiology.
3. Shier D, Butler J and Lewis R. (2003). Hole's Human Anatomy and Physiology, 10th Edition, WCB/McGraw Hill, Boston.
4. Williams S Hoar. (1991). General and Comparative Physiology, 3rd Edition, Prentice Hall of India- New Delhi.

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First Year	ALLIED PRACTICAL – II (AC)	Credit : 4
Semester - II	PLANT AND ANIMAL PHYSIOLOGY	Sub Code: 25BCAP2

(Practical)

LEARNING OBJECTIVES

- This practical course is designed to provide hands-on experience in key physiological techniques and experiments related to plant and animal systems.

PRACTICALS

1. Preparation of stained mounts of anatomy of monocot and dicot's root, stem & leaf.
2. Demonstration of opening & closing of stomata
3. Separation of photosynthetic pigments by paper chromatography.
4. Demonstration of aerobic respiration.
5. Preparation of root nodules from a leguminous plant.
6. Finding blood coagulation time
7. Determination of blood groups
8. Counting of mammalian RBCs
9. Determination of Haemoglobin.

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the principles and significance of physiological experiments in plants and animals	K1
CO2	Describe laboratory techniques used to study plant and animal physiological processes.	K2
CO3	Demonstrate experiments related to photosynthesis, transpiration, osmosis, and enzyme activity in plants and animals.	K2
CO4	Analyze physiological data obtained from experiments and interpret the results accurately.	K4
CO5	Evaluate the effects of environmental and biological factors on physiological activities through practical investigations.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

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SEMESTER III

First Year	CORE COURSE – III (CC)	Credit : 5
Semester - III	MOLECULAR BIOLOGY AND GENETICS	Sub Code: 25BTCC3

(Theory)

Objectives

- To Understand the essentials of molecular biology: replication, transcription and translation; enzymes involved in the central dogma of life, proofreading, inhibitors and post modifications.
- Thorough in prokaryotic and eukaryotic genome organization; lac& trp operon; regulation of transcription and translation in eukaryotes.

UNIT	CONTENTS	HOURS
I	Introduction to Molecular concept of gene, DNA as genetic material Experiments of Griffith; Avery, Mc Leod and Mc Carty; Harshey and Chase. RNA as genetic material- Experiment of Fraenkel and Singer. Nucleic acids: Molecular structure of DNA, Chargaff rule, Forms of DNA- A, B and Z forms. RNA types and structure-mRNA, tRNA (clover leaf model), rRNA Ribozymes.	10
II	Genome organization Fine structure of the Gene- Cistron, muton and recon. Organization of Chloroplast and mitochondrial genome. Post transcriptional modifications: methylation, polyadenylation, RNA splicing. Genetic code and its properties; process of translation- Initiation, elongation and termination. Post-translational modifications of proteins.	15
III	Gene regulation: Concept of operon, Inducible operon - Lac operon - structure and mechanism, Catabolite repression. Repressible operon Tryptophan operon structure and mechanism. Bacterial Genetics: Transformation, Transduction-Generalized and specialized; Conjugation: F factor mediated, Hfr and Sexduction.	15
IV	Transcription - Transcription factors and machinery, formation of initiation complex,transcription activator and repressor, RNA polymerases, elongation and termination, RNA processing (capping, polyadenylation, RNA editing, and splicing), RNA transport and Transcription inhibitors.	10
V	Mutation and genetic analysis of mutants: Mutation definition, types – insertion, deletion, addition, rearrangement: genetic analysis. Mutagenesis: Types: Site directed mutagenesis, base analogue mutants, tautomerization. DNA damage and repair mechanism: Photo reactivation, direct repair of nicks, excision repair, mismatch repair, recombination repair, SOS repair mechanisms.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts of molecular biology and genetics, including DNA, RNA, genes, and chromosomes.	K1
CO2	Describe the structure, organization, replication, transcription, and translation of genetic material	K2
CO3	Explain the principles of Mendelian and non-Mendelian inheritance and gene regulation mechanisms	K2
CO4	Analyze genetic variations, mutations, gene expression, and their role in heredity and evolution.	K4
CO5	Evaluate the applications of molecular biology and genetics in biotechnology, medicine, agriculture, and research.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

References and Textbooks:

- Alberts.AlexanderJohnson.JulianLewis.MartinRaff.KeithRoberts.&Peter Walter. (2008).
- Molecular Biology of theCell(5thed).GarlandScience.
- BruceAlberts, Alexander Johnson, JulianLewis, DavidMorgan, MartinRaff, Keith Roberts, Peter Walter, John Wilson, & Tim Hunt (2015). Molecular Biology of the Cell.
- 6thEdition.W.W.Norton&Company.
- Burton E. Tropp. (2012). Molecular Biology: Genes to Proteins. Jones and BartlettPublishers.GeorgeMMalacinski.(2015).Freifelder'sEssentialsof Molecular Biology.
- 4thedition.JonesandBartlettPublishers.BenjaminLewin.(2007).GenesXI. New York: Oxford University Press.David Freifelder. D. (2008). Microbial Genetics (18th ed). New Delhi: Narosa Publishing House.
- Freifelder.D.(2000).
- Molecular Biology(2nded).New Delhi:Narosa Publishinghouse.Jeyanthi,
- G.P.(2009).Molecular Biology.MJP Publishers, Chennai.Watson,J.D., Hopkins, N.H., Roberts, J.W., Steitz, J.A. & Weiner, A. M. (2013).
- MolecularBiologyoftheGene(17thed).Tokyo:The Benjamin Cummings Publishing Company Inc.
- VeerBalaRastogi.(2016).Principles of molecular biology. Medtech Publishers.
- Russel, P. (2009). Genetics:AMolecular Approach. India:PearsonEducation.

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First Year	CORE ORACTICAL – I (CP)	Credit : 4
Semester - III	LAB IN MOLECULAR BIOLOGY GENETICS	Sub Code: 25BTCP3

(Practical)

Objective:

- To Understanding about the basic techniques involved in the isolation of DNA, RNAand proteins from various sources
- In-depth analysis of DNA, protein and RNA by electrophoretic techniques
- Understanding about mitosis and meiosis in cells

PRACTICALS

1. Isolation of genomic DNA from bacteria.
2. Isolation of genomic DNA from plant.
3. Isolation of genomic DNA from Animal.
4. Separation of DNA by AGE.
5. Separation of protein by SDS-PAGE.
6. Isolation of antibiotic resistant mutant
7. Bacterial Transformation. (Demo)
8. Bacterial Conjugation. (Demo)
9. Preparation of E. coli competent cells (Demo)

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic laboratory techniques and instruments used in molecular biology and genetics experiments.	K1
CO2	Describe the principles of DNA isolation, electrophoresis, PCR, and genetic analysis techniques.	K2
CO3	Demonstrate practical skills in DNA extraction, gel electrophoresis, and other molecular biology techniques	K2
CO4	Analyze experimental results obtained from molecular and genetic studies and interpret the data accurately.	K4
CO5	Evaluate the applications of molecular biology and genetics techniques in biotechnology, medicine, and research	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference:

1. Watson JD, Baker TA, Bell SP, Gann A, Levine M and Losick R (2008) Molecular Biology of the

Gene, 6th edition, Cold Spring Harbour Laboratory Press, Pearson Publication.

2. Jeffery H. Miller 1977. Experiments in Molecular Genetics, Cold springHarbor Press.

3. N.Sivakumar (2010), Methods in Molecular Genetics, Nanjil Book stall, Nagercoil.

4. P. Oelkers, 2016. Molecular Biology Laboratory Manual

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Second Year	CORE COURSE – III (AC)	Credit : 5
Semester - III	BIOINSTRUMENTATION	Sub Code: 25BTAC1

(Theory)

Objectives

- The student should have understood the analytical techniques in the field of Biotechnology
- To make the students to understand the basic principles of Bioanalytical instruments

UNIT	CONTENTS	HOURS
I	Basics instruments: pH meter, Buffer of biological importance, Centrifuge, Laminar Air Flow Chamber, Autoclave, Hot Air Oven and Incubator.	10
II	Spectroscopic Techniques :Colorimeter, Ultraviolet and visible, Infra red and Mass Spectroscopy.	15
III	Chromatographic Techniques: Paper, Thin Layer, Column, HPLC and GC. Electrophoresis Techniques: Starch Gel, AGE, PAGE.	15
IV	Imaging techniques - Principle,Instrumentation and application of ECG, EEG, EMG, MRI, CT and PET scan radioisotopes.	10
V	Microscopy – Simple and Compound Microscope, Dark field and Light Microscope, Electron Microscope – TEM and SEM.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic principles, components, and classifications of bioinstrumentation systems.	K1
CO2	Describe the working principles and applications of instruments used in biological and biomedical sciences.	K2
CO3	Explain the operation and functions of analytical instruments such as spectrophotometers, microscopes, centrifuges, and electrophoresis systems.	K2
CO4	Analyze the role of bioinstrumentation in biological research, clinical diagnosis, and biotechnology applications.	K4
CO5	Evaluate the advantages, limitations, and selection criteria of bioinstrumentation techniques for specific applications.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

References and Textbooks:

- Introduction Practical Biochemistry–S.K. Sawhney and Randhir Singh. Narosa Publishing House

- Principles of Applied Biomedical Instrumentation-Gedder A and L.E. Balsar, John Wiley and Sons.

- Modern Experimental Biochemistry 2ndEdition-Boyer,Rodney F.Benjamin and Cummins

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First Year	ALLIED PRACTICAL – III (AP)	Credit : 4
Semester - III	LAB IN BIOINSTRUMENTATION	Sub Code: 25BTAP1

(Practical)

Objective:

- To introduce an fundamentals of cell biology and microbiology techniques
- To teach students the basic techniques and instrument principles in biotechnology
- To give hands on microbiology and cell biology experiments

PRACTICALS

1. Laboratory rules and regulations
2. Handling of Microscopy
3. Preparation of Buffer
4. Seperation of biomolecules by Thin layer Chromotography
5. Demonstration of Electrophoresis Techniques
6. Paper Chromotography
7. Study and Operation of Centrifuge
8. Demonstration of ELISA

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the principles, components, and applications of bioinstrumentation used in biological and biomedical laboratories.	K1
CO2	Describe the working principles of instruments such as pH meter, colorimeter, spectrophotometer, centrifuge, and microscope.	K2
CO3	Demonstrate the operation, calibration, and maintenance of common bioinstrumentation devices.	K2
CO4	Analyze experimental data obtained using bioinstrumentation techniques and interpret the results accurately.	K4
CO5	Evaluate the suitability and performance of different bioinstrumentation tools for biological research and diagnostic applications.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

CO3	Apply bioinformatics tools for sequence alignment, database searching, and molecular analysis	K2
CO4	Analyze genomic, proteomic, and phylogenetic data using bioinformatics approaches.	K4
CO5	Evaluate the role of bioinformatics in genomics, proteomics, drug discovery, and biotechnology research.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

References and Textbooks:

1. Attwood.T.K. Parry D.J. and Smith. Introduction to Bioinformatics. A joint Publications. 1999

2. Baldi, P. and Brunak, S. Bioinformatics: The Machine Learning Approach. MIT Press. 2001

3. A.D. Baxevanis and B.F.F. Ouellette (Eds). Bioinformatics: A practical guide to the analysis of genes and proteins. John Wiley and Sons. 2002

4. Gentleman, R. Bioinformatics and Computational Biology Solutions using R and Bioconductor. Springer Science and Business media Inc. 2005

5. http://www.biologydiscussion.com/biodiversity/bioinformatics/notes-on bioinformatics-genetics/38224
6. http://www.biotechnologynotes.com/bioinformatics/notes-on-bioinformatics/693 29

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SEMESTER IV

Second Year	CORE COURSE – IV (CC)	Credit : 5
Semester - IV	IMMUNOLOGY	Sub Code: 25BTCC4

(Theory)

Objectives

- This course presents the basic defense mechanism of animals
- To make the student to understood the concept immunology
- On successful completion of the subject the student should have understood: Immunity Antigen, Antibody, Cell of immune system and their function and regulations

UNIT	CONTENTS	HOURS
I	Immunology- Introduction, Historical perspective to the immune system.Innate immunity and acquired immunity.Primary and secondary lymphoid organs-structure and functions.Cells of the immune system, Antigens, Immune responses.	10
II	Immuno globulin- structure and functions, Immunoglobulin classes and biological activities.Epitopes - Properties of B-cell and T- cellepitopes. Monoclonal anti bodies Production, Classical uses.	15
III	Immune cells -T- cells-Receptors, Maturation, Activation and differentiation. Cell mediated immune response, Cytokines and cytokine receptors. B- cells- Activation of B- cells by T-cell independent antigen and T-cell dependent antigens. Antibody mediated immune responses, Differentiation of B-cells.	15
IV	Major His to compatibility complex-MHC, HLA complex, HLA typing. Immunological tolerance, auto immunity and autoimmune diseases.	10
V	Immuno hematology - ABO blood grouping, Rh Blood grouping, and medical applications of blood grouping, blood transfusion and hemolytic disease of the new born.Hyper sensitivity Type I, II, III, IV and V.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic concepts, components, and functions of the immune system.	K1

CO2	Describe the structure and functions of innate and adaptive immunity, immune cells, and lymphoid organs.	**K2**
CO3	Explain antigen-antibody interactions, immune responses, and mechanisms of immune regulation.	**K2**
CO4	Analyze the immunological basis of hypersensitivity, autoimmunity, transplantation, and immunodeficiency disorders.	**K4**
CO5	Evaluate the applications of immunology in disease diagnosis, vaccine development, and immunotherapy.	**K3**
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

References and Textbooks:

- Immunology-Dulsy Fatima and N.Arumugam-Saras Publication.
- Immunology (7th Edition)-David Male, Jonathan Brost off, David Roth, Ivan Roitt, Mosby. Elsevier
- Immunology-A short course-Eli Benjamin, Geoffrey Suashine, Sidney Leskowitz. John wiley&sons.Inc. Publications.
- Fundamental Immunology (6th Edition)- William E Paul, Walters Kluwer Hedth, Lippincott William Wilkins.
- https://www.researchgate.net/publication/313343876_Immunology_Lecture_Notes_Immune_responses
- https://www.easybiologyclass.com/tag/immunology-short-notes/ 19

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Second Year	CORE PRACTICAL – IV (CP)	Credit : 4
Semester - IV	IMMUNOLOGY	Sub Code: 25BTCP4

(Practical)

Objective:

- Enrich the practical skills on the immunological techniques
- Understand the immune reactions, antigen-antibody reactions, separation of lymphocytes, purify immunoglobulins and detect antigens via techniques like western blotting

PRACTICALS

1. ABO Blood Grouping and Rh Factor typing
2. RBC and WBC Count by Hemo cytometers.
3. Blood cell Analysis
4. Measurement of pulse and blood pressure.
5. Precipitation reactions (Radial immuno or Double immuno diffusion test)
6. Preparation of serum & plasma from blood samples
7. Separation of lymphocytes from whole blood

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Recall the basic concepts of immunology, laboratory safety practices, and immunological techniques.	K1
CO2	Explain the principles of antigen-antibody interactions and various immunological assays.	K2
CO3	Perform immunological experiments such as blood grouping, agglutination, precipitation, and ELISA techniques.	K2
CO4	Analyze and interpret the results obtained from immunological tests and experiments.	K4
CO5	Evaluate the significance and applications of immunological techniques in clinical diagnosis and biomedical research	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

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Second Year	CORE COURSE – IV (CC)	Credit : 5
Semester - IV	BIOINFORMATICS	Sub Code: 25BTAC2

(Theory)

Objectives

- The course aims to provide the students with an experimental and computational knowledge to embrace a systems biology approach and experience authentic systems genetics research by designing and conducting independent research projects.

UNIT	CONTENTS	HOURS
I	Introduction about bioinformatics and biological databases BioInformatics: Introduction, definition, objectives and scope. Application of BioInformatics. General Introduction of Biological Databases; Nucleic acid databases (NCBI, DDBJ, and EMBL), Protein databases	10
II	Sequence analysis- Introduction to Sequences, alignments and Dynamic Programming; Pairwise alignment (BLAST and FASTA Algorithm) and multiple sequence alignment (Clustal W algorithm) and phylogenetic analysis.	15
III	Similarity search, gene identification and prediction -Similarity Search Introduction, Working with FASTA, Working With BLAST.Basis of gene prediction, Pattern Recognition, Gene prediction Methods, Other gene prediction tools.	15
IV	Protein classification and structure visualisation-Overview of the Protein Structure, Protein Structure Visualization and prediction: Pymol, Rasmol, and Structure - based Protein Classification, Protein Structure databases, Protein Structure Visualization Databases	10
V	Protein structure prediction-Protein Identification and Characterization, Primary structure analysis and Prediction, Secondary Structure Analysis and Prediction, Protein Function Prediction and Protein Prediction from a DNA Sequence., Homology modeling	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Recall the fundamental concepts, scope, and importance of bioinformatics in biological sciences.	K1
CO2	Explain the organization and applications of biological databases and information retrieval systems.	K2

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Second Year	ALLIED PRACTICAL – IV (AP)	Credit : 4
Semester - IV	BIOINFORMATICS(P)	Sub Code: 25BTAP2

(Practical)

Objective:

- To organize data,to access existing information and to submit new entries as they are produced.
- To develop tools and resources that aid in the analysis of data
- To analyze the data and interpret the results in a biologically meaningful manner

PRACTICALS

1. Retrieval of Nucleotide sequence from GenBank
2. Retrieval of Protein sequence from GenBank
3. Sequence Similarity Search using BLASTN
5. Sequence Similarity Search using BLASTP
6. Accessing Structural Database and Download the Protein Structure
7. Working with EMBL
8. Multiple Sequence Alignment
9. Predicting Physiochemical properties of protein sequence
10. Predicting Peptide mass of protein sequence
11. Predicting cleavages of protein sequence

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Recall the fundamental concepts, scope, and importance of bioinformatics in biological sciences.	K1
CO2	Explain the organization and applications of biological databases and information retrieval systems.	K2
CO3	Apply bioinformatics tools for sequence alignment, database searching, and molecular analysis	K2
CO4	Analyze genomic, proteomic, and phylogenetic data using bioinformatics approaches.	K4
CO5	Evaluate the role of bioinformatics in genomics, proteomics, drug discovery, and biotechnology research.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

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SEMESTER V

Third Year	CORE COURSE – V (CC)	Credit : 5
Semester - V	GENE MANIPULATION TECHNOLOGY	Sub Code:25BTCC5

(Theory)

Objectives

- To understand the fundamental principles and cloning vectors used in genetic engineering.
- To gain knowledge on tools and techniques used in genetic engineering
- To understand the concept of gene cloning and construction of gene library

UNIT	CONTENTS	HOURS
I	Overview of Restriction and Modification system. Overview of Gene Cloning; Cloning vehicles: Plasmids, Single strand DNA vector – M13 Phage. Cosmids, Plasmids, BAC and YAC. Expression vector –RNA probe synthesis, High level expression of proteins, Protein solubilization, purification and export. Application of Gene Cloning – DNA analysis in research and Biotechnology.	10
II	DNA library – Types and importance. cDNA library, Chromosome walking. Differences between gDNA and cDNA library. Screening strategies – Hybridization, PCR, Immuno screening. Overview on microarray and its applications, Microarrays – Principles ofDNA microarray technology, steps and techniques involved, types of DNA arrays, applications, advantages and disadvantages	15
III	DNA sequencing – Importance, methodology, Chemical & Enzymatic methods, Pyrosequencing, How to sequence a genome? , Short gun sequencing and clone contig approach, Automated sequence, Genome sequencing methods – top down approach, bottom up approach.	15
IV	PCR – Principle and applications. Different types of PCR – Hot start PCR, Touchdown PCR, Multiplex PCR, Inverse PCR, Nested PCR, AFLP-PCR, Allele specific PCR, Assembly PCR, Asymmetric PCR, LATE-PCR, Colony PCR, in-situ PCR, Long P CR. Real-time PCR SYBR Green assay, Taqman Probes, Molecular beacons.	10
V	Introduction to foreign genes into animal cells –Micro injection, Retroviral vectors, Trasnsfection of Embryonic stem cells, recombination. Transgenic plants – Importance Ti Plasmid. Overview of Gene therapy. Gene therapy – Introduction and types. Methods of Gene therapy, target sites, Gene targeting and silencing, Transgenic plants and its applications	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic concepts, principles, and applications of gene manipulation technology and genetic engineering.	K1
CO2	Explain the structure and function of DNA, genes, vectors, enzymes, and recombinant DNA technology.	K2
CO3	Describe the methods used in gene cloning, DNA amplification (PCR), gene transfer, and sequencing techniques.	K2
CO4	Analyze the mechanisms of gene expression, regulation, and genetic modification in prokaryotic and eukaryotic systems.	K4
CO5	Apply knowledge of recombinant DNA technology in agriculture, medicine, industry, and biotechnology research.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

References and Textbooks:

- Glick,B.R.,andPasternak,J.J.(2018).MolecularBiotechnology: Principles and Applications of Recombinant DNA. 5th edition. Washington, DC: ASM Press.
- Reinert,J.,andBajaj,Y.P.S.(1997).AppliedandFundamentalAspects of Plant Cell, Tissue and Organ Culture. Berlin: Springer.
- Sambrook, J., Fritsch, E.F., and Maniatis, T. (1989). Molecular Cloning: ALaboratoryManual.2ndedition.ColdSpringHarbor,NY:ColdSpring Harbor Laboratory Press.
- Gahlawat,S.K.,Duhan,J.S.,Salar,R.K.,Siwach,P.,Kumar,S.,&Kaur, P.(2018).AdvancesinAnimalBiotechnologyanditsApplications. Springer.

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Third Year	CORE COURSE – VI (CC)	Credit : 5
Semester - V	PLANT AND ANIMAL BIOTECHNOLOGY	Sub Code: 25BTCC6

(Theory)

Objectives

- To Understoodusage of Plant andAnimal products and exploitation ofthem in Biotechnology.
- KnowledgeonCropdevelopmentandCallusculture
- Biotechnologicalapplicationsofplants,Animaltissueculture

UNIT	CONTENTS	HOURS
I	Cell and Tissue culture Introduction to cell and tissue culture, Plant tissueculture media, plant hormones and growth regulators in tissue culture, Preparation of suitable explants for organogenesis. Micropropagationon large scale, somatic embryogenesis, protoplast culture and somatichybridization, Anther,pollenandovaryculture forproductionofhaploid plants.	10
II	Cell culture methods .Cell culture methods for the secondary metabolite production, somaclonal variation and its significance, Cryopreservation, Plant transformationtechniques-Agrobacteriummediatedgenetransfer– Mechanism ofDNAtransfer, generalfeaturesofTIandRIplasmidsandtheiruseasvectors, role of virulence genes, reporter genes, use of 35S and other promoters, reporter genes.	15
III	Animal cell cultures .Animal cell cultures: Culture media – composition and preparation, Role of CO2, serum and protein-free defined media and their applications; Culturing and maintenance of different animal cell lines. Characterization of cultured cell, cyto – toxicity and growthparameters.Stemcellcultures,embryonicstemcellandtheirapplicatio n, cell culturebased vaccines, apoptosis.	15
IV	TransgenicanimalsTransgenicanimals:Methodofobtainingtransgenicanim als using fertilized eggs and embryonic blastocyst cell, importance of transgenic animals – increased productivity of domestic animals, improved desired characters of domestic animals, production of recombinant gene products and proteins for pharmaceutical use. Animal models for tackling human diseases (Gene knock out in mice models).	10
V	Animal cloning .Transgenic silkworms, Animal cloning: Methods of cloning in animalsystem – Rat, Sheep, Pig; importance ofcloning – Gene therapyand cell mediated therapy. Ethical issues in Animal	10

	Biotechnology.	

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the principles, scope, history, and applications of plant and animal biotechnology.	K1
CO2	Explain cell culture techniques, genetic engineering methods, and recombinant DNA technology used in plants and animals.	K2
CO3	Apply biotechnology techniques for crop improvement, animal breeding, disease diagnosis, and vaccine production.	K2
CO4	Analyze the role of molecular markers, transgenic organisms, tissue culture, and cloning in biotechnology	K4
CO5	Evaluate the ethical, environmental, social, and commercial implications of plant and animal biotechnology applications.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

References and Textbooks:

- Plant genetic engineering, DoddsJ.H.
- Plant molecule biology, GriersonandS.V.Convey
- Molecular biotechnology, Principle and applications of recombinant DNA technology,
- Bernard R.Glick.
- Plant Biotechnology-MonicaHughes.
- Related Online Contents[MOOC,SWAYAM]

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Third Year	CORE COURSE – VII (CC)	Credit : 5
Semester - V	MICROBIAL AND DAIRY BIOTECHNOLOGY	Sub Code: 25BTCC7

(Theory)

Objectives

- Understand the growth properties and kinetics of the industrially important microorganisms
- Study the different types of fermenters and their design
- Acquire knowledge of the upstream and downstream processes of fermentation

UNIT	CONTENTS	HOURS
I	Gene cloning of industrial microorganisms: Gene cloning – strategies in gene cloning – Importance, advantage and methods of gene cloning in Bacillus, Pseudomonus, Streptomyces and yeast.	10
II	Selection of industrial strains and media formulation: Isolation and screening of Industrially important microbes, strain improvement, media formulation and sterilization.	15
III	Bioreactors: Bioreactor design, parts and their functions. Bioprocess control and monitoring variables such as temperature, agitation, pressure and pH. Computers in bioprocess control system, its types. Introduction to large scale production of recombinant proteins (insulin, Biopolymer) using bioreactors.	15
IV	Microflora of raw milk - sources of contamination. Spoilage and preservation of milk and milk products. -antimicrobial systems in raw milk.Importanceofbiofilms,theirroleintransmissionofpathogens in dairy products and preventive strategies.	10
V	Food fermentations: Indian Pickles Bread, vinegar,fermented vegetables (sauerkraut), fermented dairy products (yoghurt, cheese, Acidophilus Milk, Kefir, Koumiss).	10

CO#	COURSE OUTCOMES	K-LEVELS

CO1	Define the principles of microbial and dairy biotechnology and explain their importance in food and dairy industries.	**K1**
CO2	Describe the characteristics of industrially important microorganisms and their applications in dairy fermentation processes.	**K2**
CO3	Explain microbial growth, metabolism, fermentation techniques, and their role in the production of dairy products.	**K2**
CO4	Analyze the use of biotechnology tools, genetic engineering, and microbial cultures in improving dairy products and processes.	**K4**
CO5	Apply quality control, safety standards, and microbial testing methods in dairy biotechnology and processing industries.	**K3**
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

References and Textbooks:

- Joanne M.Willey, Linda M.Sherwood and ChristopherJ.Woolverton– Prescott's Microbiology
- K.Jayaram Paniker andAnanthanarayan– Text book of Microbiology
- MichaelJ. Pelczar, E.C.S.ChanandNoelR.Krieg– Microbiology
- Michael T.Madigan, John M. Martinko and Jack Parker – Brock Biology of Microorganisms
- World Health Organization (WHO)–Laboratory Biosafety Manual

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Third Year	CORE COURSE – V (CP)	Credit : 4
Semester - V	GENE MANIPULATION TECHNOLOGY PLANT AND ANIMALBIOTECHNOLOGY MICROBIAL DAIRY BIOTECHNOLOGY(P)	Sub Code: 25BTCP5

Objective:

- Developing the basic skills to search of DNA, Proteindatabases.
- Predict the structure of DNA&proteins.
- To give hands on experience in immunological techniques and Plant Tissue Culture
- The basic techniques of microbial biotechnology.
- Developing the skill for handling industrially important microbes and maintenance.

PRACTICALS

1. Isolation of plasmid DNA from Bacteria
2. Analysis of plasmid DNA in Agarose Gel electrophoresis
3. Isolation of DNA from animal tissues.
4. Demonstration of Southern Hybridization, Synthesis of cDNA.
5. Plant Tissue Culture Media Preparation
6. Invitro germination of seeds
7. Callus induction and differentiation
8. Embryo Culture (demo)
9. Isolation and fusion of protoplast
10. Artificial seed production
11. Micropropagation
12. Qualitative analysis of alkaloids, flavonoids, saponins, tannins and phenolic compound
13. Maintenance of pure cultures of industrially important microbes
14. Fermentor, Productionmedia- Principles&applications
15. Alcohol fermentation by yeast,and quantification of ethanol
16. Citric acid fermentation by Aspergillus

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Identify and demonstrate laboratory techniques, instruments, and safety practices used in gene manipulation, animal, plant, and dairy biotechnology.	K1
CO2	Perform DNA isolation, gel electrophoresis, and basic recombinant DNA techniques used in gene manipulation studies	K2
CO3	Demonstrate animal cell culture and plant tissue culture techniques under aseptic laboratory conditions.	K2
CO4	Apply microbial and fermentation techniques for the preparation, analysis, and quality assessment of dairy	K4
CO5	Analyze and interpret experimental results obtained from gene manipulation, animal and plant biotechnology, and dairy biotechnology experiments.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

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- Plant Biochemistry and Molecular Biology by P.J. Lea and R.C. Leegood (1993)JohnWiley&Sons.3.Plants,GenesandAgriculturebyM.J.Chrispeels and D.F. Sadava (1994) Jones Jones and Barlett
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- BiochemistryandMolecularBiologyofPlantsbyBuchanan,B.B.W. Gruissen and AndR.L.Jones (2000), American Society of Plant Biology, Rockilled, Md, USA.
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- AppliedPlantBiotechnologybyIgnacimuthu,S.,(1996)TataMcGrawhill.
- AgriculturalBiotechnologybyPurohit,S.S.,(2003)AgrobiosIndia.
- PlantMolecularBiologyandBiotechnology,A.Slater,N.ScottandM.Flower (2003) and Oxford University Press, Oxford

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Third Year	MAJOR BASED ELECTIVE –I (MBE)	Credit : 4
Semester - V	A. PHARMACEUTICAL BIOTECHNOLOGY	Sub Code: 25BTMB1A

(Theory)

Objectives

- Students will understand the series of processes involved in drug development, patenting and drug approval.
- Will learn about Biopharmaceuticals
- Will become familiar with Biotechproteindrugs
- Will understand aboutmanagement of drugs
- Will be familiar with Pharmaceutical sectors

UNIT	CONTENTS	HOURS
I	ObjectivesofPharmaceuticalBiotechnology-GenericandBiogenericdrugs. Stages in the drug development process -Drug discovery - Drug designing - Drug production - Preclinical trials - Clinical trials - Pharmacokinetics and Pharmacodynamics - Patenting & Drug Approval - Drug Marketing - Post clinical trials.	10
II	Production of recombinant proteins - Development of Nucleic acid based therapies-Biopharmaceuticalconsiderations - Pharmaceuticalregulations - Formulation of Biotechnology products - Drug delivery – Pharmacognosy	15
III	HumanInsulin(Humulin).Growthhormones(Humatrope)-Bloodcoagulating factor (factor VIII - Kogenate) - Erythropoietin- (Epogen) Granulocyte colony stimulating factors (Neulasta) - Interferons (Avonex) - Antimicrobial peptides (β - defensin 2) - Vaccines (Pentavac). Biologics (Humira - Adalimumab). - Cancer based biologics (rituximab).	15
IV	Drug toxicity analysis - Common side effects of drugs and managements – Drugsofabuse-Lifechangingcomplications - Preventionandmanagement	10

V	National and International Drug approval agencies - Top National and Internationalpharmaceuticalindustries - Scopeandcareeropportunitiesin pharmaceutical sectors.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the concepts, scope, and applications of pharmaceutical biotechnology in healthcare and drug dev	K1
CO2	Describe the principles and production of biopharmaceuticals, including recombinant proteins, vaccines	K2
CO3	Explain the role of genetic engineering, molecular diagnostics, and gene therapy in pharmaceutical biotechnology	K2
CO4	Analyze biotechnological methods used in drug discovery, development, formulation, and large-scale manufacturing of pharmaceuticals.	K4
CO5	Evaluate the quality assurance, regulatory requirements, biosafety, and ethical aspects associated with biopharmaceutical products.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference

1. Chandrakant Kokateand Pramod H.J1stEdition(2011),Text Book of Pharmaceutical Biotechnology, Elsevier

2. Crommelin,DeanJ.A., Sindelar, Robert, Meobohm, Bernd(Eds.) (2019), Pharmaceutical Biotechnology: Fundementals and Applications, Springer.

3. AshishDixit, Pawan Tiwariand Vivekanand Kishan Chatap (2015), Textbook of Pharmaceutical Biotechnology, Studium Press (India) Pvt. Ltd.

4. John F. Corpenter, Mark C. Manning (2012). Rational Design of stable formulation Theory and Practice,(1stedition),US: Springer Science, ISBN: 9781461351313.

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Third Year	MAJOR BASED ELECTIVE –I (MBE)	Credit : 4
Semester - V	B.MEDICAL BIOTECHNOLOGY	Sub Code: 25BTMB1B

(Theory)

Objectives

- Understand broad theoretical knowledge and critical understanding of advanced principles in antibodies and vaccines
- Gain the practical knowledge required to support a career in biomedical research environment
- Have a sound plat form for setting up basic therapeutical production
- Gain the knowledge about clinical trials and applications in the knowledge about the diagnostics of infectious disease

UNIT	CONTENTS	HOURS
I	Antibodies and vaccines - Therapeutic production of antibodies, antibody mediated drugdeliveryofvaccines, different kind ofvaccines and applications ofrecombinantvaccines.Diagnosis - Biochemicaldiagnostics,inbornerrorsof metabolism, haemoglobinopathies.	10
II	Molecular basis of disease, Recombinant DNA Technology in medicine, gene probes as molecular diagnostic reagents. Polymerase Chain Reaction in clinical diagnostics, DNA sequencing of representative clones to detect mutations	15
III	Diagnosis of infectious diseases, Viral diseases –HIV, influenza; bacterial diseases-enteric diseases, mycobacterium diseases; immune arrays. FACs immune histochemical staining, ELISA, FISH techniques.	15
IV	Immuno blot analysis of antigen sand allergens. Production of therapeutic agents – Productions and application of therapeutic agents, Production of cytokines and interferons	10
V	Principles of project management in clinical trials and its application. Principles of research ethics; Ethical issues in clinical trials; Animal rights and use of animals in the advancement of medical technology. Use of humans in Scientific Experiments.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Evaluate the quality assurance, regulatory requirements, biosafety, and ethical aspects associated with biopharmaceutical products.	K1
CO2	Describe the molecular and cellular basis of diseases and the role of biotechnology in their diagnosis and treatment	K2
CO3	Explain the principles and applications of recombinant DNA technology, gene therapy, stem cell technology	K2
CO4	Analyze the use of biotechnological tools in the development of vaccines, therapeutic proteins, monoclonal antibodies, and personalized medicine.	K4
CO5	Evaluate the ethical, legal, regulatory, and biosafety issues associated with medical biotechnology and its clinical applications.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference

- Roli,M.(2017).National Ethical Guidelines for Biomedical and Health Research Involving Human Participants, ISBN: 978-81-910091-94
- Lela, B. and Maribeth, L. F. (2011). Molecular Diagnostics: Fundamentals, Methods and Clinical Applications, (1st Edition) . Philadelphia, USA. FADavisCompany.ISBN-13:978-0803626775
- Clinical Applications,(1stEdition). Philadelphia, USA. FADavis Company. ISBN-13: 978-0803626775

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Third Year	SKILL BASED ELECTIVE –I (SBE)	Credit : 2
Semester - V	A. MEDICAL LAB TECHNOLOGY	Sub Code: 25BTSB/A

(Theory)

Objectives

- To understand the basic aspects of medical biotechnology, pathogenesis of human diseases, disease diagnosis, management, drug discovery, development and Clinical research.
- To provide an overview of genetic diseases and the diagnostic techniques used in the medical field.
- To provide knowledge on antigens, antibodies and theories of antibody formation.

UNIT	CONTENTS	HOURS
I	General Anatomy - Cell-structure & function. Tissue - Epithelium, Connective, Sclerous muscle. Systemic anatomy - Basic features of Cardio vascular system, Respiratory system, Digestive system, Excretory system, Reproductive (Male &Female) system, Nervous system	10
II	Blood -Blood cells, Haemoglobin, Blood grouping, Coagulation Factors. HaematologicalDisorders-Classification of Anaemia-Iron Deficiency anaemia, Megaloblastic anaemia, HaemolyticAnaemia.	15
III	Cardio vascular system-Heart rate, cardiac cycle, cardiac output, blood pressure, hypertension, radial pulse.Cardiac Profile-Description, Hypertension, Angina, Myocardial Infarction, Pattern of Cardiac Enzymes in heart diseases.Respiratory System-Ventilation, functions, Lungs- Volumes and capacities.	15
IV	Gastrointestinal System - Process of digestion. Endocrinology - Endocrine Glands Description. Hormones-their secretion and functions. Excretory system, Structure of nephron, Urine formation.	10
V	Histopathology -Introduction. Basic steps for tissue processing-Fixing, Embedding, Microtomy, Staining, Mounting, methods of decalcifications.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic concepts and principles of Medical Laboratory Technology and its role in healthcare.	K1
CO2	Describe laboratory procedures and diagnostic techniques used in hematology, microbiology, biochemistry, and p	K2
CO3	Apply standard laboratory methods for specimen collection, processing, and analysis of clinical samples.	K2
CO4	Analyze laboratory findings and correlate results for disease diagnosis and patient care.	K4
CO5	Demonstrate professional skills in handling laboratory instruments, maintaining quality control, and following ethic	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference

- RobbinsS.L.(1974)PathologiclbasisofDisease.WBSaundersCompany
- GuytonA.C. andHallJ.E. (2006)TextbookofMedicalPhysiology11th edn. Saunders
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Third Year	SKILL BASED ELECTIVE –I (SBE)	Credit : 2
Semester - V	B.MEDICAL CODING	Sub Code: 25BTSBMB

(Theory)

Objectives

- Convert medical diagnoses, procedures, and services into standardized alphanumeric codes accurately.
- Ensure proper documentation for billing and insurance claims.
- Maintain compliance with healthcare regulations and coding guidelines.
- Reduce claim denials and improve reimburs

UNIT	CONTENTS	HOURS
I	Introduction to US Healthcare.Healthcare in India and US . Patient, Provider and Payers Relationship Importance and significance ofcoding intoday's world	10
II	Anatomy,TerminologyandPathophysiologyTerminology–Rootword, suffix, prefixes, combining forms. Human body – Anatomical Position, Body CavitiesDermatology OphthalmologyOtorhinolaryngologyCardiology GastroenterologyGenitourinarysystemGynaecologyandObstetricsOrthopaedicsEndocrinology Immunology, Lymphatics, and Hematology.	15
III	ICD-9-CM. History, Volumes. Conventions. General Coding Guidelines . Chapter Specific Guidelines Infections and Parasitic Diseases. Neoplasms . Endocrine, Nutritional and metabolic diseases, and immunity disorders. Diseases of blood and blood forming organs. Mental disorders Diseases of Nervous system Diseases of Sense organs. Diseases of Circulatory System.	15
IV	Current Procedural Terminology. Introduction to CPT. CategoryI to Category III . Six sections of CPT in Detail-Evaluation and Management Surgery Radiology Medicine Significance of Parent codes. CPT Modifiers Symbols and significance Alphanumeric codes overview of categories II and III	10

V	HCPCS (Healthcare Common Procedure Coding System) Introduction to HCPCS Significance and Usage .Types of HCPCs . Modifiers Level II HCPCS	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts of medical coding and healthcare documentation systems.	K1
CO2	Describe international coding systems such as ICD, CPT, and HCPCS used in healthcare services.	K2
CO3	Apply coding principles and assign appropriate diagnostic and procedural codes from medical records.	K2
CO4	Analyze patient documentation and coding data to ensure accurate billing and reimbursement processes.	K4
CO5	Demonstrate professional skills in maintaining coding accuracy, compliance standards, and ethical practices	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference

- CarolJ.Buck–Step-by-StepMedicalCoding
- AmericanMedicalAssociation(AMA)–CurrentProceduralTerminology (CPT)
- Elsevier–ICD-9-CMExpertforPhysicians
- DorothyT.Iozzio–MedicalTerminology,AnatomyandPhysiology
- World HealthOrganization(WHO)–InternationalClassificationofDiseases (ICD)
- PaulineC.Jorwekar–MedicalBillingandCodingEssentials
- AnnEhrlichandCarolL.Schroeder –MedicalTerminologyfor Health Professions

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SEMESTER VI

Third Year	CORE COURSE –VIII (CC)	Credit : 5
Semester - VI	ENVIRONMENTAL BIOTECHNOLOGY	Sub Code: 25BTCC8

(Theory)

Objectives

- The course empowers students to Acquire skills in bioremediation of environmental pollutants
- Apply the skills in developing innovative biotechnological processes for waste
- conversion, resource recovery, and production of bioproducts bioresources.
- Knowledge in environmental biotechnology for gene cloning

UNIT	CONTENTS	HOURS
I	Ecosystem - structure, functions. Energy flow and mineral cycle-C, N, P. Environmental problems -Ozone depletion, Green House Effect, Water, Air, Soil pollution, Land degradation.	10
II	Waste water-Physical, Chemical and biological characteristics. Introduction to water microbiology, Water borne diseases. Waste water treatment - Physical, Chemical and biological. Membrane filtration and Reverse Osmosis. Waste water treatment efficiency assessment.	15
III	Xenobiotics - Xenobiotic compounds, Biodegradation of xenobiotics, Biological detoxification, Hazardous waste management. Mining and metal biotechnology-Copper and Iron, Microbial transformation, accumulation and concentration of metals, Metal leaching extraction.	15
IV	Bio fuels - Introduction, Production of non-conventional fuels-Methane (Biogas), Hydrogen, Alcohols and algal hydrocarbons. Use of microorganisms in augmentation of petroleum recovery	10
V	Environmental Genetics- Degradation plasmids (TOL), Release of genetically engineered microbes (GEM) in the environment, Impact of GEM in environment, Role of GEM in degradation of industrial pollutants. Bio sensors and microprobes.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the basic concepts and principles of Environmental Biotechnology and its applications in environmental manag	K1
CO2	Describe the role of microorganisms and biotechnological processes in pollution control and waste management.	K2
CO3	Apply biotechnological techniques for environmental monitoring, bioremediation, and wastewater treatment.	K2
CO4	Analyze environmental issues and evaluate biotechnological approaches for sustainable environmental managemen	K4
CO5	Demonstrate knowledge and practical skills related to environmental conservation, pollution prevention, and eco-friendly	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference

- Bernaral R. Glick & Jack J. Pastemak. (1994). Molecular Biotechnology:
- Principles and Applications
- of Recombinant DNA. ASM Press. Washington, DC USA.
- Brown, T.A. (1995). Gene cloning - A introduction. Chapman & Hall, London. Buckley, R. G. (2019). Environmental Microbiology. CBS Publication.
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- Neetu Sharma, Abhinashi Singh Sodhi, & Navneet Batra. (2021). Basic Concepts in Environmental Biotechnology. CRC Press.
- Rittmann, E.B., & Perry, L. (2020). Environmental Biotechnology: Principles and Applications. 2nd Edition. McGraw Hill Publication.

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Third Year	CORE COURSE- X (CC)	Credit : 5
Semester - VI	BIOSAFETY, BIOETHICS, AND IPR	Sub Code: 25BTCC9

(Theory)

Objectives

- Imply the Applications of statistics in biological data
- Differentanalyticalmethodsforbiologicaldata
- Microbialriskmanagementandsafetyanalysis
- Developknowledgeofintellectualpropertyrights
- Knowaboutpatentapplicationprocess
- EmphasizethepurposeofethicalregulationsonBiotechnologyand bioethical issues

UNIT	CONTENTS	HOURS
I	Biosafety - Introduction, biosafety issues in biotechnology, historical background. Primary Containment for Biohazards, Biosafety Levels, Biosafety Levels of Specific Microorganisms.	10
II	Biosafety Guidelines - Biosafety guidelines and regulations - National and International. Definition of GMOs & LMOs. Environmental release of GMOs, Risk Analysis, Assessment, management and communication.	15
III	Bioethics - What is bioethics - legal and socioeconomic impacts of biotechnology, Public education of the process of biotechnology, making ethical concerns of biotechnology research and innovation	15
IV	Intellectual Property Rights - Intellectual property rights - TRIPS, GATT, International conventions patents and methods of application of patents - Legal implications - Biodiversity and farmer rights	10
V	Patents and patent laws - Objectives of the patent system, Basic principles and general requirements of patent law. Biotechnological inventions and patent law. Legal development - Patentable subjects and protection in biotechnology. The patenting living organisms.	10

Second Year	CORE PRACTICAL – VI (CP)	Credit : 5
Semester - VI	ENVIRONMENTAL BIOTECHNOLOGY	Sub Code: 25BTCP6

(Practical)

Objective:

- To provide sound knowledge about ecosystem, bioremediation and metal mining.
- The course content aims to make the Students understand how biotechnology can help in monitoring or removing the pollutants and developing an understanding of new trends such as biofuels, renewable energy sources, or microbial technologies which can minimize the harmful impact of pollutants in the environment.
- To make students understand the environmental crisis and about its control measures.

PRACTICALS

1.Estimation of total solids in industrial wastes

2. Analysis of TDS in industrial wastes

3. Estimation of suspended solids in effluent water

4. Determination of COD and BOD in sewage water

5. Analysis of Microbial contamination in air by open plate method

6. Isolation of phage from sewage.

7. Preparation of phage stock.

8. Estimation of total hardness in water.
9. Estimation of nitrite in water sample.
10. Estimation of salinity in water samples.

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Identify and understand laboratory instruments, safety measures, and basic techniques used in Environmental B	

CO2	Describe methods used for the analysis of environmental samples such as water, soil, and wastewater.	
CO3	Perform experiments related to microbial isolation, environmental monitoring, bioremediation, and waste tre	
CO4	Analyze experimental data and interpret results related to environmental pollution and biotechnological applications	
CO5	Demonstrate practical skills in handling laboratory equipment, maintaining records, and applying environmental biotechnological techniques for sustainable management.Define the fundamental concepts and principles of Nanobiotechnology and its interdisciplinary applications.	
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Text Books:

- Scragg A. H, Environmental Biotechnology, (2nd revised edition), Oxford University Press 2005
- Jogdand S. N, Environmental Biotechnology (3rd edition), Himalaya publishing house pvt.ltd 2012.
- Thakur. I. S, Environmental Biotechnology: Basic Concepts and Applications, (2nd revised edition), I K International Publishing House Pvt. Ltd, 2011.

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Third Year	MAJOR BASED ELECTIVE –II (MBE)	Credit : 4
Semester - VI	A. NANO BIOTECHNOLOGY	Sub Code: 25BTMB2A

(Theory)

Objectives

- To know about nanoparticles and their analysis using Advanced Instrumentation
- To get an insight about Nanodevices
- The students will know about the Applications of Nanobiotechnology
- The students will know about the Nano Biosensors and their applications

UNIT	CONTENTS	HOURS
I	Glimpse of Nanotechnology based material in ancient India: Wootz steel (iron carbide) and the Delhi iron pillar (anticorrosive nanomaterial), Bhasma (nanomaterial as medicine). Contributions of Indian Research Institutes in the field of nanobiotechnology	10
II	Metals: Silver nanoparticle synthesis and its analyses by UV-spectroscopy and FTIR. Self-Assembly nanomaterial: Cell membrane and its analyses by SEM	15
III	Nano-thin films: Chitosan thin film, Nanodevices (nanorobots), Nanotubes: Microtubules assembly and its importance, Nano shells- Dendrimers: Liposomes, Nanofibers: Collagen, Fibronectin & elastin, nano fluidics: Extracellular matrix assembly and its importance	15
IV	Agriculture: Crop production- Nano fertilizers technology, Biomaterial to improve shelf life of vegetables. Medicine: Collagen thin films in wound healing mechanism, Nanoscale devices – DNA microarray for disease diagnosis, Antibodies and Targeted drug delivery system.	10
V	Nano biosensors (Firefly-luciferase) and its applications, Introduction to Biomimetics (Gecko foot effect, Lotus leaf effect: Paint and fabrics, Box fish based Car).	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts and principles of Nanobiotechnology and its interdisciplinary applications.	K1
CO2	Describe the synthesis, properties, and characterization of nanomaterials used in biological systems.	K2
CO3	Apply nanobiotechnology techniques in drug delivery, diagnostics, biosensors, and therapeutic applications.	K2
CO4	Analyze the interactions, advantages, and potential risks of nanomaterials in biological and environmental systems.	K4
CO5	Demonstrate knowledge and practical skills related to the applications of nanobiotechnology in healthcare, agriculture, a	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference

- D.Voet&J.G.Voet (2010), Biochemistry, John Wiley &Sons, New York.
- Biochemistry by Lubert Stryer, 4th Ed., WH.Freeman, 1995.
- David S. Goodsell, "Bionanotechnology", John Wiley &Sons Inc., publications, 2004.
- Guozhong Cao (2004). Nanostructures and Nanomaterials, synthesis, properties and applications, Imperial College Press, ISBN: 978-1860944802.
- 5.C.M.Niemeyer, C.A. Mirkin (2007). Nanobiotechnology, WILEY-VCH Verlag GmbH & Co. KG, Weinheim, ISBN: 9783527306589

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Third Year	MAJOR BASED ELECTIVE –II (MBE)	Credit : 4
Semester - VI	B. FOOD BIOTECHNOLOGY	Sub Code: 25BTSB2B

(Theory)

Objectives

- This is helps to the students to understand about food preparation
- To know about Food borne diseases.
- To help the students to understand about the industrial food process

UNIT	CONTENTS	HOURS
I	Biotechnology relating to the food industry – Role of bioprocess engineering in biotechnology industry- Regulatory and social aspects of biotechnology in foods- Application of biotechnology in waste treatment of food industries. Historical evolution of food processing technology.	10
II	Cereals and Millets. Wheat- composition, types (hard, soft/ strong, weak). Malting, gelatinization of starch. Rice- and composition, parboiling of rice- advantages and disadvantages. Composition of pulses, toxic constituents in pulses, processing of pulses soaking, germination, decortications, cooking and fermentation. Fats and Oils. Refining of oils, Rancidity.	15
III	Classification of fruits and vegetables, general composition, enzymatic browning, names and sources of pigments, Dietary fibre. Post-harvest changes in fruits and vegetables –physical changes, chemical changes, pathological changes during the storage of fruits and vegetables.	15
IV	Concept of red meat and white meat, composition of meat, marbling, post mortem changes in meat- rigor mortis, tenderization of meat, ageing of meat. Aquaculture, composition of fish, characteristics of fresh fish, spoilage of fish - microbiological, physiological and biochemical. Composition and nutritive value of egg, deterioration of egg quality. Milk and Milk Products. Chemical composition of milk, its constituents, processing of milk, pasteurization, homogenization.	10
V	Types of food adulterants – test to detect adulterants in foods – metal contaminants - contaminants of processed foods- Food products as analytical samples, general aspects of biosensors- biosensors for food contaminant analysis, commercially available biosensors for food analysis. Food additivies, FSSAI regulations, Methods of fortifying and enriching food	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts and principles of Food Biotechnology and its significance in food production.	K1
CO2	Describe the role of microorganisms, enzymes, and fermentation technologies in food processing and preservation.	K2
CO3	Apply biotechnological techniques for enhancing food quality, nutritional value, and safety.	K2
CO4	Analyze the applications of biotechnology in food preservation, functional foods, and value-added product development.	K4
CO5	Demonstrate knowledge and practical skills related to food quality assurance, safety standards, and industrial applications of food biotechnology.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference

- D.G. Rao, 2010. Fundamentals of Food Engineering –PHI Learning Pvt Ltd.
- Narang, Food Microbiology
- Michael P. Doyle, Larry. R. Food Microbiology – Fundamentals & Frontiers
- Frazier, Food Microbiology
- Yiu Hui & G. Khachatourians, Food Biotechnology
- Ibek, Laramie & Bhunia, Fundamentals of Food Microbiology, CRC Press.

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Third Year	SKILL BASED ELECTIVE –II (MBE)	Credit : 2
Semester - VI	A. MOLECULAR DIAGNOSTICS	Sub Code: 25BTSB2A

(Theory)

Objectives

- Theobjectiveofthecourse istomakeStudentsawareofthevarious medical diagnostic techniques and their use in diagnosing various disorders in humans.

UNIT	CONTENTS	HOURS
I	Genetics and diagnostics: General features of Chromosomes, chromosome banding patters, banding techniques and their correlates, karyotyping, DNA profiling hybridization arrays. Early detection of diseases.	10
II	Molecular methods: Nucleic acid extraction: principles and methods. Assessing purityandconcentrationofnucleicacids,PCR-basicandapplied-Alu-PCR,Hot start PCR, PCR-ELISA, Arbitrarily primed PCR, in situ PCR.	15
III	Nuclear hybridization methods, Single nucleotide polymorphisms and plasmid fingerprintingininfections,PFGE, DGGE. Detectionofmutationusing ARMS-PCR and microsatellite markers.	15
IV	Allele susceptibility test formultifactorial disorders (Neuraltubedefect,cleft-lip and palate, cardiovascular disorder, male infertility). Diagnosis of inbornerrors.	10
V	Cell sorting- Flow cytometry and FACS. Neonatal and prenatal diagnosis. Sex identification in forensics.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts and principles of Molecular Diagnostics and its applications in disease diagnosis.	K1
CO2	Describe molecular techniques such as PCR, RT-PCR, DNA sequencing, and nucleic acid hybridization used in diagnostics.	K2
CO3	Apply molecular diagnostic methods for the detection and identification of genetic and infectious diseases.	K2

CO4	Analyze molecular diagnostic data and interpret results for clinical and research applications.	**K4**
CO5	Demonstrate knowledge and practical skills related to molecular diagnostic tools, laboratory procedures, and quality assurance practices.	**K3**
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference

- Wilson, K.andWalker,J.PracticalBiochemistry–Principlesandtechniques 7th edition, 2010, Cambridge University Press.
- PrimroseSandyB.andRichardTwyman,PrinciplesofGeneManipulation and Genomics (7th Edition), Wiley-Blackwell 2006.
- BrownT. A, GeneCloningandDNAAnalysis:AnIntroduction, (6thEdition) Wiley-Blackwell, 2010.

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Third Year	SKILL BASED ELECTIVE –II (MBE)	Credit : 2
Semester - VI	B.STEM CELL TECHNOLOGY	Sub Code: 25BTSB28

(Theory)

Objectives

- The objective of this paper is to familiarize the students with stem cell technologyanditsapplicationsforbettermentofthesociety.Thecourseis designed to give a broad view of mammalianstemcells, reviewingwhere they are found in the body, the different types and how they are cultured.
- The topics will cover the basic biology of these stem cells as well as bioengineering and application of these stem cells to potential treatments of human diseases.

UNIT	CONTENTS	HOURS
I	Introduction to stem cells Definition, properties, proliferation, culture of stem cells, medicalapplicationsofstemcells, ethicaland legalissues inuseofstem cells.	10
II	Types ofstemcells, StemCellbiologyand therapy, types embryonic stem cell, Adult stemcell, Stem Cell Biology and Therapy,EmbryonicStemCells,culture and the potential benefits of stem cell technology	15
III	Therapeuticapplicationsofstemcells, GeneTherapy:Introduction, Historyand evolution of Gene therapy, optimal disease targets, Failures andsuccesses with gene therapy and future prospects	15
IV	GeneticPerspectivesforGene Therapy,GeneDeliverymethods:Viralvectors and Non-viral Vectors	10
V	Ethical Issues associated with stem cell-based regenerative medicine field Regulatory and Ethical Considerations of stem cell and Gene Therapy, AssessingHumanStemCellSafety, Use ofGeneticallyModifiedStemCells in Experimental Gene Therapies.	10

CO#	COURSE OUTCOMES	K-LEVELS
CO1	Define the fundamental concepts, types, and characteristics of stem cells and their biological significance.	K1
CO2	Describe the principles of stem cell proliferation, differentiation, and their role in development and tissue regeneration.	K2
CO3	Apply stem cell biology concepts in regenerative medicine, disease modeling, and therapeutic applications.	K2
CO4	Analyze the role of stem cells in tissue repair, disease treatment, and biomedical research.	K4
CO5	Demonstrate knowledge and practical understanding of stem cell biology, ethical considerations, and research applications.	K3
K1-Remember; K2 -Understand; K3-Apply; K4-Analyze; K5-Evaluate ; K6-Create		

Reference

- StemCellBiologyandGeneTherapy. QuesenberryPJ, SteinGS,eds. (£65.00.) Wiley, 1998.
- Progressingenetherapy,Volume2,Pioneeringstem cell/genetherapytrials, Roger Bertolotti, Keiya
- OzawaandH.KirkHammond,VSPinternationalsciencepublishers
- StemCellsHandbook:StewartSell,HumanaPress;TotowaNJ,USA;Oct.2003,
- HumanEmbryonicStemCells:ThePracticalHandbookbyStephenSullivanand Chad A Cowan.

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NME

SECOND YEAR	NON MAJOR ELECTIVE- I	COURSE CODE:
SEMESTER III	A. FUNDAMENTALS OF INFORMATION TECHNOLOGY	25CSNM1A CREDITS: 2

COURSE OBJECTIVES:

- To familiarize the students with the world of IT and IT-enabled services.
- To provide a in-depth knowledge about internet and internet tools.
- To enable the students to understand about Computer Security

UNITS	CONTENTS	HOURS
I	Introduction to Computers-Generation of Computers-Classification of Digital Computer - Anatomy of Digital Computer.	6
II	CPU and Memory-Secondary Story Devices-Input Devices-Output Devices.	6
III	Introduction to Computer Software-Programming Language-Operating Systems -Introduction to Database Management System.	6
IV	Computer Networks-WWW and Internet-Email-Web Design.	6
V	Computers at Home, Education, Entertainment, Science, Medicine and Engineering - Introduction to Computer Security - Computer Viruses, Bombs, Worms.	6

TEXTBOOKS:

1. **Alexis Leon & Mathews Leon** (2019), *Fundamentals of Information Technology*, Leon Tech World, Chennai.

REFERENCE BOOKS:

1. **V. Rajaraman** (2018), *Introduction to Information Technology*, PHI Learning Pvt. Ltd., New Delhi.
2. **P.K. Sinha & Priti Sinha** (2017), *Computer Fundamentals*, BPB Publications, New Delhi.
3. **V. Rajaraman & Neeharika Adabala** (2015), *Fundamentals of Computers*, PHI Learning Pvt. Ltd., New Delhi.

WEB RESOURCES:

1. Computer Fundamentals Tutorial – TutorialsPoint
2. Information Technology and Computer Basics – GeeksforGeeks
3. Computer Science and IT Resources – IBM SkillsBuild

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SECOND YEAR	NON MAJOR ELECTIVE I	COURSE CODE: 25CSNM1B
SEMESTER III	B.BASICS OF OFFICE AUTOMATION	CREDITS: 2

COURSE OBJECTIVES:

- To create the professional-quality documents.
- To create dynamic slide presentations with animation digitally and effectively.
- To expose the students with organize and analyze information in worksheet.

UNITS	CONTENTS	HOURS
I	Introduction to Microsoft Office Navigate the Microsoft Office command structure using the Ribbon- Quick Access Toolbar Minimum Toolbar and Shortcut Menus - Use Microsoft Office help.	6
II	Ms. Word: Format text- paragraphs and document elements-Identify and use correct sequencing and spacing of all required business letter components- Format a research paper with footnotes and bibliographical list of sources-Insert a Word table, enter data and format the table- Streamline document creation using templates Insert and format a graphic or word art in Word documents.	6
III	Ms. Excel: Enter text-numbers and formulas into a worksheet applying to business situations- Create and edit Excel charts to graphically present numerical data- Format a worksheet using number formats, borders, color, fonts and conditional formats- Use the Excel SUM, AVERAGE, MAX, MIN, IF, and NOW functions to display desired calculations- Create and copy formulas using relative and absolute cell references- Edit worksheets and add new worksheets to a workbook, rename and rearrange sheets.	6
IV	Ms. Access Create a database file consisting of at least two tables using proper designing principles- Create a report that presents the data in the database in an organized and flexible manner- Sort the records of a database and use criteria to search for specific information- Create queries to retrieve specific information from one or more related tables.	6
V	Ms. PowerPoint Create single and multi-level bulleted list slides- Use the PowerPoint ribbon to change Layouts and Themes in presentations- Insert and size clip art, photographs, and shapes into a presentation- Create and run a presentation with slide transitions	6

SECOND YEAR	NON MAJOR ELECTIVE II	COURSE CODE: 25CSNM2A
SEMESTER IV	A:WORKING PRINCIPLES OF INTERNET	CREDITS: 2

COURSE OBJECTIVES:

- To teach the basics of the World Wide Web
- To Understand the fundamentals of the Internet and the usage
- To know the components of Multimedia on the internet

UNITS	CONTENTS	HOURS
I	What is Internet? The Internet's underlying Architecture	6
II	Connecting to the Internet– Communicating on the Internet	6
III	How the World Wide Web works .Common Internet tools	6
IV	Multimedia on the Internet–Intranet and shopping on the Internet	6
V	Safeguarding the Internet	6

TEXTBOOKS:

- Internet & World Wide Web: How to Program **Authors:** Harvey Deitel and Paul Deitel **Publisher:** Pearson Education Covers Internet fundamentals, web technologies, web communication, and Internet applications.

- The Internet Book: **Author:** Douglas Comer Provides a clear introduction to Internet architecture, protocols, and networking concepts.

REFERENCE BOOKS:

1. PrestonGralla,"HowtheInternetWorks",PearsonEducation,EighthEdition,2006.
2. C.Xavier, Fundamentals of Internet and Emerging Technologies, New Age International
3. AlexisLeon,InternetforEveryone,S.Chand(G/L)&CompanyLtd;SecondEdition2012.
4. AndreaC.Nakaya,"InternetandSocialMediaAddiction",ReferencePointPress,2015.

SECOND YEAR	NON MAJOR ELECTIVE- II	COURSE CODE: 25CSNM2B
SEMESTER IV	B. FUNDAMENTALS OF WEB DESIGNING	CREDITS: 2

COURSE OBJECTIVES:

- To understand the basic concepts of computer networks, client–server architecture, and web technologies.
- To learn Internet fundamentals including IP addressing, domain names, and communication protocols (HTTP, FTP, SMTP, etc.).
- To gain knowledge of web browsing, search engines, email systems, and web navigation techniques.
- To design web pages using HTML including text formatting, lists, images, links, and tables.
- To understand XML fundamentals and differentiate between HTML .

UNITS	CONTENTS	HOURS
I	Introduction to Web related terms :Client –Server–Single Tier Architecture–Two Tier Architecture– N Tier Architecture – Internet – Intranet – WWW – Web Page – Web Site.	6
II	Introduction to Internet and its Protocols: Internet Addressing – IP Address – Domain Name – Electronic Mail–Uniform Resource Locator –TCP/IP-FTP–HTTP–SHTTP-SMTP-POP–UDP – PPP – Telnet – Gopher – WAIS.	6
III	SearchingtheWeb:Browser:Web Index–WebSearch Engine–Meta Search Sites-Directoryand Indexes – Managing Mails: Address Book – File Attachment Facility – Junk Mail.	6
IV	Introduction to HTML: HTML Code for Web Page–HTML Tags-Web Page Basics–Display a Web Page in a Web Browser – Format texts – Font Style and sizes – Lists – Ordered Lists – Unordered Lists – Image – Add an image – Background Image – Horizontal Rule	6
V	Introduction to HTML Table: <TABLE> - <TH> - <TD> - <TR> - Cell Padding – Cell Spacing – Table Borders.	6

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Nagore Post, Nagapattinam – 611 002, Tamilnadu.

PART - I				
SEM	**PART**	**COURSE**	**TITLE**	**SUBJECT CODE**
I	I	Language Course - I (LC) - Tamil	Tamil - I	25LCT1
II	I	Language Course - II (LC) - Tamil	Tamil - II	25LCT2
III	I	Language Course - III (LC) - Tamil	Tamil - III	25LCT3
IV	I	Language Course - IV (LC) - Tamil	Tamil - IV	25LCT4
SEM	**PART**	**COURSE**	**TITLE**	**SUBJECT CODE**
I	I	Language Course - I (LC) - French -	French - I	25LCF1
II	I	Language Course - I (LC) - French -	French - II	25LCF2
III	I	Language Course - I (LC) - French -	French - III	25LCF3
IV	I	Language Course - I (LC) - French -	French - IV	25LCF4
SEM	**PART**	**COURSE**	**TITLE**	**SUBJECT CODE**
I	I	Language Course - I (LC) - Arabic	Arabic - I	25LCA1
II	I	Language Course - I (LC) - Arabic	Arabic - II	25LCA2
III	I	Language Course - I (LC) - Arabic	Arabic - III	25LCA3
IV	I	Language Course - I (LC) - Arabic	Arabic - IV	25LCA4
SEM	**PART**	**COURSE**	**TITLE**	**SUBJECT CODE**
I	I	Language Course - I (LC) - Hindi	Hindi - I	25LCH1
II	I	Language Course - I (LC) - Hindi	Hindi - II	25LCH2
III	I	Language Course - I (LC) - Hindi	Hindi - III	25LCH3
IV	I	Language Course - I (LC) - Hindi	Hindi - IV	25LCH4
SEM	**PART**	**COURSE**	**TITLE**	**SUBJECT CODE**
I	I	Language Course - I (LC) - Sanskrit	Sanskrit - I	25LCS1
II	I	Language Course - I (LC) - Sanskrit	Sanskrit - II	25LCS2
III	I	Language Course - I (LC) - Sanskrit	Sanskrit - III	25LCS3
IV	I	Language Course - I (LC) - Sanskrit	Sanskrit - IV	25LCS4

PART - II				
SEM	**PART**	**COURSE**	**TITLE**	**SUBJECT CODE**
I	II	English Language Course - I (ELC)	English - I	25ELC1
II	II	English Language Course - II (ELC)	English - II	25ELC2
III	II	English Language Course - III (ELC)	English - III	25ELC3
IV	II	English Language Course - IV (ELC)	English - IV	25ELC4
Common Papers				
SEM	**PART**	**COURSE**	**TITLE**	**SUBJECT CODE**
I	IV	Value Education	Value Education	25VEUGC1
II	IV	Environmental Studies	Environmental Studies	25ESUGC2
III	IV	Health and Wellness	Health and Well	25HWUGC3
V	IV	Soft Skills Development	Soft Skills Development	25SSUGC4
VI	V	Gender Studies	Gender Studies	25GSUGC5
VI	V	Extension Activities	Extension Activities	

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B.Sc ~~Biochemistry~~ Biotechnology

SEM	PART	COURSE	TITLE	Course	SUBJECT CODE
I	III	Core Course – I (CC)	Cell Biology	CC1	25BTCC1
	III	Core Practical – I (CP)	Cell Biology (P)	CP1	25BTCP1
	III	Allied Course – I (AC)	Biochemistry	AC1	25BCAC1
	III	Allied Practical – I(AP)	Biochemistry (P)	AP1	25BCAP1
II	III	Core Course – II (CC)	Microbiology	CC2	25BTCC2
	III	Core Practical – II (CP)	Microbiology (P)	CP2	25BTCP2
	III	Allied Course – II (AC)	Plant and Animal Physiology	AC2	25BCAC2
	III	Allied Practical-II (AP)	Plant and Animal Physiology (P)	AP2	25BCAP2
III	III	Core Course – III (CC)	Molecular Biology and Genetic Engineering	CC3	25BTCC3
	III	Core Practical – III (CP)	Molecular Biology and Genetic Engineering (P)	CP3	25BTCP3
	III	Allied Course – III (AC)	Bioinstrumentation	AC3	25BTAC1
	III	Allied Practical – III(AP)	Bioinstrumentation (P)	AP3	25BTAP1
IV	III	Core Course – IV (CC)	Immunology	CC4	25BTCC4
	III	Core Practical – IV(CP)	Immunology (P)	CP4	25BTCP4
	III	Allied Course – IV (AC)	Bioinformatics	AC4	25BTAC2
	III	Allied Practical – IV(AP)	Bio informatics (P)	AP4	25BTAP2
V	III	Core Course – V (CC)	Gene Manipulation Technology	CC5	25BTCC5
	III	Core Course – VI (CC)	Plant and Animal Biotechnology	CC6	25BTCC6
	III	Core Course – VII (CC)	Microbial and Dairy Biotechnology	CC7	25BTCC7
	III	Core Practical – V(CP)	Gene Manipulation Technology, Plant and Animal Biotechnology, Microbial and Dairy Biotechnology	CP5	25BTCP5
	III	Major based Elective –I	A. Pharmaceutical Biotechnology	MBE1	25BTMB1A
	III		B. Medical Biotechnology	MBE1	25BTMB1B
	III	Internship	Internship	BTIS	25BTIS
VI	III	Core Course – VIII (CC)	Environmental Biotechnology	CC8	25BTCC8
	III	Core Course – X (CC)	Biosafety, Bio ethics and IPR	CC9	25BTCC9
	III	Core Practical – VI (CP)	Environmental Biotechnology	CP6	25BTCP6
	III	Major Based Elective –II	A. Nano Biotechnology	MBE2	25BTMB2A
	III		B. Food Biotechnology	MBE2	25BTMB2B
	III	Project	Dissertation and Viva-voce	PROJECT	25BTPW

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IT	5	IV	SBE-I	Amazon Web Services	25CSSB1A
	5	IV	SBE-I	Web Technology	25ITSB1B
	6	IV	SBE-II	Cyber Security	25CSSB2A
	6	IV	SBE-II	Office Automation	25ITSB2B
CS	5	IV	SBE-I	Amazon Web Services	25CSSB1A
	5	IV	SBE-I	Web Designing	25CSSB1B
	6	IV	SBE-II	Cyber Security	25CSSB2A
	6	IV	SBE-II	Enterprise Resource Planning	25CSSB2B
MATHS	4	IV	SBE-I	Introduction to Latex	25MASB1
	5	IV	SBE-II	Computational Mathematics	25MASB2
Chemistry	5	IV	SBE-I	Cosmetics and Personal care Products	25CHSB1A
	5	IV	SBE-I	Food Chemistry	25CHSB1B
	6	IV	SBE-II	Dyeing Techniques and water Treatment	25CHSB2A
	6	IV	SBE-II	Forensic Science	25CHSB2B
Physics	5	IV	SBE-I	Electrical wiring fundamentals	25PHSB1A
	5	IV	SBE-I	Home appliances maintenance and servicing	25PHSB1B
	6	IV	SBE-II (P)	Domestic electrical appliances and measuring instruments	25PHSB2
	6	IV	SBE-II		
Bioche	5	IV	SBE-I	Biomedical Instumentation	25BCSB1A
	5	IV	SBE-I	Basics of forensic science	25BCSB1B
	6	IV	SBE-II	MedicalLab Technology	25BCSB2A
	6	IV	SBE-II	Medical coding	25BCSB2B
Biotech	5	IV	SBE-I	Medical Lab Technology	25BTSB1A
	5	IV	SBE-I	Medical Coding	25BTSB1B
	6	IV	SBE-II	Molecular Diagnostics	25BTSB2A
	6	IV	SBE-II	Stem Cell technology	25BTSB2B
NAD	5	IV	SBE-I	Traditional Herbs in Food Science	25NDSB1A
	5	IV	SBE-I	Human Development and Counseling	25NDSB1B
	6	IV	SBE-II	Textile Science and Clothing	25NDSB2A
	6	IV	SBE-II	Principles of Interior Designing	25NDSB2B
ViscoM	5	IV	SBE-I	Cultural Tourism in India	25VCSB1A
	5	IV	SBE-I	Tourism Product	25VCSB1B
	6	IV	SBE-II	Corporate Communication	25VCSB2A
	6	IV	SBE-II	Event Management	25VCSB2B
	5	IV	SBE-I	Export Analysis & Documentation	25FTSB1A
	5	IV	SBE-I	Apparel Production & Planning	25FTSB1B

	4	IV	NME-II	Web Design using HTML	25ITNM2B
CS	3	IV	NME-I	Fundamentals of Information Technology	25CSNM1A
	3	IV	NME-I	Basics of Office Automation	25CSNM1B
	4	IV	NME-II	Working principles of Internet	25CSNM2A
	4	IV	NME-II	Web Designing	25CSNM2B
MATHS	3	IV	NME-I	Quantitative Aptitude I	25MANM1A
	3	IV	NME-I	Computational Mathematics I	25MANM1B
	4	IV	NME-II	Quantitative Aptitude II	25MANM2A
	4	IV	NME-II	Computational Mathematics II	25MANM2B
Chemistry	3	IV	NME-I	Drugs and Cosmetics	25CHNM1A
	3	IV	NME-I	Chemistry in Everyday life	25CHNM1B
	4	IV	NME-II	Textile Chemistry	25CHNM2A
	4	IV	NME-II	Dairy Chemistry	25CHNM2B
Physics	3	IV	NME-I	Introduction to Digital electronics	25PHNM1A
	3	IV	NME-I	Principle of Electronics	25PHNM1B
	4	IV	NME-II	Laser physics	25PHNM2A
	4	IV	NME-II	Medical physics	25PHNM2B
Bioche	3	IV	NME-I	Medicinal Diet	25BCNM1A
	3	IV	NME-I	Health & Nutrition	25BCNM1B
	4	IV	NME-II	First aid &Home Nursing	25BCNM2A
	4	IV	NME-II	Life style disease	25BCNM2B
Biotech	3	IV	NME-I	Food and Nutritional Biotechnology	25BTNM1A
	3	IV	NME-I	Herbal Biotechnology	25BTNM1B
	4	IV	NME-II	Biotechnology for Human Health	25BTNM2A
	4	IV	NME-II	Dairy Biotechnology	25BTNM2B
NAD	3	IV	NME-I	Diet in Human Health	25NDNM1A
	3	IV	NME-I	Nutrition for Everyday Life	25NDNM1B
	4	IV	NME-II	Diet Counseling and Education	25NDNM2A
	4	IV	NME-II	Nutrition for Health and Fitness	25NDNM2B
	3	IV	NME-I	Basics of Communication	25VCNM1A